MIL-R-22184B(SHIPS)
8 November 1965
SUPERSEDING
MIL-R-2218A(SHIPS)
11 December 1962
(See 6.3)

MILITARY SPECIFICATION

RECEIVING SET, LORAN C AN/SPN-38( ) AND AN/WPN-5( )

1. SCOPE

1.1 Scope.- This specification governs the design, development, manufacture, and performance testing of a marine navigation receiving set which will operate in the Loran system, type C (Loran C). It is intended that the circuitry for providing the necessary functions for furnishing data readout be kept to an absolute minimum consistent with performance as required by this specification. Although the specified form factor and weight are not considered unduly restrictive, the design with respect to these parameters is eased by the use of a single phase 115-v., 400 cycle primary source of power and the encouraged utilization of transistors and other miniaturized types of construction where permissible and practical.

1.2 Purpose.- It is intended that the equipment procured under this specification be completely compatible with the system performance capability of Loran C. Loran C operates on a carrier frequency of 100 kilocycles (kc.) with a utilization bandwidth of 20 kc. It employs automatic envelope and cycle synchronous detectors and post-detector narrow banding techniques to produce highly accurate navigational information, even under conditions of signal-to-noise ratios of less than unity, from positions on the system baseline to distances in the order of 2,000 miles. The degree of permissible deterioration of accuracy due to speeds encountered in surface ships up to 45 knots, and submarines up to 30 knots, and the requirement for holding rough synchronization during acceleration due to turning at such speeds are included in the design problems. Special attention must also be given to the suppression of strong interfering radio signals which exist adjacent to and in the Loran C frequency band. The sensitivity of the radio receiver of this equipment must be high in order to maximize the utility of this long range system.

FSC 5825

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1.3 Equipment to be furnished.- One complete Loran navigation receiving set of the type specified (see 6.1) shall consist of the following:

(a) Surface ship type, AN/SPN-38
  (1) One Loran receiver, R-1275/SPN-38( ).
  (2) One whip antenna coupler unit, CU-1342/SPN-38( ).
  (3) One whip antenna, AT-1011/U( )
  (4) One set special tools, as required.[1]

(b) Submarine type, AN/WPN-5.
  (1) One Loran receiver, R-1276/WPN-5( ).
  (2) One loop and trailing wire antenna preamplifier and coupler, CU-1343/WPN-5( ).
  (3) One set special tools, as required.[2]

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-17/7 - Cable, Radio Frequency, Coaxial, RG-12A/U.
MIL-C-17/10 - Cable, Radio Frequency, Coaxial, RG-17A/U.
MIL-S-901 - Shock Tests, H.I.(High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-T-16784 - Terminal Boards.
MIL-I-16910 - Interference Measurement, Electromagnetic, Methods and Limits.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.
MIL-M-23313 - Maintainability Requirements for Shipboard and Shore Electronic Equipment and Systems.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw Type, and Associated Terminal Board Lugs, General Specification for.

[1]Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

[2]Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

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STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment Design Standard for.

DRAWINGS

BUREAU OF SHIPS

RE49B740 - RG-264A/U Cable.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 Definitions.-

3.1.1 Loran C is a long-range precision Loran system employing automatic envelope and radio frequency (r.f.) cycle comparison techniques. It operates in the frequency band 90-110 kc. with a carrier frequency of 100 kc. It utilizes group pulsing (eight pulses/group at each slave station and nine pulses/group at the master station) techniques in lieu of single-pulsed transmissions

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in order to raise the average transmitted power. The pulses of the group are phase-coded (carrier frequency cycle phase relationship with respect to envelope is changed from pulse to pulse in a group in accordance with a prescribed code). The use of synchronous detection and post-detector narrow banding reduces the effect of noise and interference and in conjunction with the use of phase coding virtually eliminates the effects of skywave contamination. In addition, phase-coding provides signal identification required for automatic search operation. A Loran C chain consisting of one master and two or three slave stations operates on a single specific repetition rate.

3.1.2 Basic repetion rate.- The basic repetition rate is the approximate rate at which a Loran transmitter is single or group pulsed. There are six basic rates, as follows:

SS - 10 pulses per second (p.p.s.).
SL - 12-1/2 p.p.s.
SH - 16-2/3 p.p.s.
S - 20 p.p.s.
L - 25 p.p.s.
H 33-1/3 p.p.s.

3.1.3 Specific repetition rate.- The specific repetition rate is the precise rate at which a Loran transmitter is single or group pulsed. Specific rates are identified by a letter followed by a numeral. The letter signifies the basic repetition rate, and the numeral indicates the number of hundreds of microseconds (usec.) that the specific period is less than the basic period. Eight specific rates numbered 0 to 7, inclusive, are provided. Repetition periods in microseconds for each of the six basic and eight specific rates are tabulated below:

Specific rate	SS	SL	SH	S	L	H
0	100,000	80,000	60,000	50,000	40,000	30,000
1	99,900	79,900	59,900	49,900	39,900	29,900
2	99,800	79,800	59,800	49,800	39,800	29,800
3	99,700	79,700	59,700	49,700	39,700	29,700
4	99,600	79,600	59,600	49,600	39,600	29,600
5	99,500	79,500	59,500	49,500	39,500	29,500
6	99,400	79,400	59,400	49,400	39,400	29,400
7	99,300	79,300	59,300	49,300	39,300	29,300

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3.1.4 Receiver r.f. selectivity.- The Loran receiver r.f. selectivity characteristic is the receiver property that determines discrimination between a signal on the frequency of greatest sensitivity and an undesired signal on any other frequency. The response of any particular frequency is expressed as the inverse ratio of an AØ signal voltage on that frequency to an AØ voltage whose frequency is that at which receiver response is a maximum when both produce the same r.f. output as viewed on the oscilloscope.

3.1.5 Filter signal rejection.- Filter signal rejection is the ratio of (a), the AØ signal input voltage required to give a reference oscilloscope amplitude when the filter is adjusted to a minimum output; to (b), the AØ signal voltage at the same frequency required to give the same reference oscilloscope amplitude without filter inserted.

3.1.6 Signal-to-noise ratio.- For the purpose of this specification signal-to-noise is defined to be the square root of the ratio of the power of a pure CW signal having the same peak-to-peak amplitude as the peak-to-peak amplitude of the r.f. cycle at the pulse standard sampling point (30 usec. after the beginning of the pulse) to the power of a white noise signal as measured at the output of the receiver prior to detection. A signal-to-noise ratio of 1:1 can be established in the following manner: With an oscilloscope and a non-reactive, linear, power measuring meter connected at the r.f. output of the receiver, a peak-to-peak CW signal is fed into the receiver and set for suitable deflection on the oscilloscope. The CW signal power is noted on the meter. The CW signal is removed and the peak-to-peak value of the r.f. cycle at the sampling point of a Loran signal input is adjusted to obtain the same deflection on the oscilloscope. The Loran input is then removed and a suitable noise source input is adjusted to give the same power reading as the CW signal. When the noise and Loran source are both connected, a signal to noise (s/n) ratio of 1:1 is obtained as defined herein. Other s/n ratios shall be obtained by reducing the noise or signal input by means of an accurately calibrated resistive attenuator.

3.1.7 Standard sampling point.- All measurements in the Loran C system are normally referred to a standard sampling point which is defined as being the point 20, 25 or 30 usec (adjustable within the equipment) after the beginning of the pulse. Unless otherwise specified the value for the standard sampling point is 25 usec.

3.1.8 Loran C signal strength.- Loran C signal strength is specified in terms of the root mean square (r.m.s.) value of a continuous-wave signal having the same r.f. amplitude as the r.f. cycle at the standard sampling point.

3.1.9 Loran C master station (M).- A Loran C master station is one of the group of Loran C stations comprising a single Loran rate. It has the burden of precisely maintaining its assigned repetition rate and of monitoring the synchronization accuracy and stability of its paired slave stations.

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3.1.10 Loran slave station(W,X,Y,Z).- A Loran slave station is the Loran transmitting station whose transmissions paired with those of its master station comprise a Loran pair. It has the burden of maintaining an accurately fixed delay between the reception of the master signal and transmission of slave signals.

3.1.11 Time delay (time difference).- Time delay (time difference) is a measure of the relative times of reception of master and slave signals. It is measured as the time in microseconds between the reception of the master signal and reception of the slave signal.

3.1.12 Loran C pulse group.- A Loran C pulse group consists of eight phase-coded pulses transmitted by any one Loran C station. These eight pulses are separated by 500 or 1000 usec. within the group. However it is required that this equipment will normally utilize only the 500 usec. spacing. Selection of the 1000 usec. spacing shall be a behind-the-panel operator control.

3.1.13 Loran C group repetition rates.- The Loran C group repetition rates correspond to the specific repetition rates under 3.1.3.

3.1.14 Multipulse triggers.- Multipulse triggers are a series of pulses generated within the receiver which correspond to the Loran C r.f. pulses transmitted during one pulse group repetition interval. Eight triggers are generated during each transmission (master and slaves) in every repetition interval.

3.1.15 Phase coding.- Phase coding is a method of changing the r.f. cycles relative to the envelope of each pulse in the Loran C pulse group. The phase is varied from pulse to pulse in accordance with a prescribed program.

3.1.16 Blinking.- Blinking is a method by which the transmitting stations may indicate a malfunction in system timing accuracy or to communicate other information to system users.

3.1.17 Signal to interference ratio.- Signal to interference ratio is defined as the square root of the ratio of the power of a pure continuous wave signal having the same peak-to-peak amplitude as that of the r.f. cycle at the standard sampling point, to the power of the interfering signal.

3.2 Material.- Material and parts shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

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3.3 General requirements.- The equipment shall meet the requirements of MIL-E-16400 except as otherwise specified herein. The design objectives, as specified in MIL-E-16400, for simplicity, reliability, ease of maintenance and installation consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.3.1 Temperature.- The receiver and submarine antenna preamplifier and coupling units shall meet the requirement for class 4 temperature conditions as specified in MIL-E-16400. The whip antenna and coupling unit shall meet the requirements for class 2 temperature conditions as specified in MIL-E-16400.

3.3.2 Size.- The size of each unit and the total volume of the equipment shall be kept to a minimum consistent with practical design. The total volume shall not exceed 4.0 cubic feet and shall include all units. The equipment dimensions shall be compatible with the requirements of MIL-E-16400 for submarine and surface ship installations. The equipment form factor shall be submitted to the bureau or agency concerned for approval.

3.3.3 Weight.- The weight of the navigation receiver (exclusive of whip antenna) shall not exceed 175 pounds. The whip antenna shall not exceed 30 pounds.

3.3.4 Power requirements.-

3.3.4.1 Primary power source.- The equipment shall be designed to operate from a primary power source of 115 volts, 400 cycles, single phase.

3.3.4.2 Regulation.- To provide for variations in line voltage, separate transformer taps shall be provided for line voltages of 105, 115, and 125 volts. The equipment shall be so designed as to operate satisfactorily with a variation of $\pm$ 5 percent without recalibration or adjustment.

3.3.4.3 Power source transients.- Equipment performance during transients of supply voltage and frequency shall conform to MIL-E-16400.

3.3.4.4 Power input.- The total input power required by the equipment will be held to a practical minimum, and shall not exceed 335 watts.

3.3.5 Radio frequency interference.- Radio frequency interference reduction and elimination shall be in accordance with MIL-I-16910. Elimination of harmful effects caused by external radiated and conducted interfering signals shall be accomplished as part of the equipment basic design as a matter of good engineering practice.

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3.3.6 Degree of enclosure.- The degree of enclosure shall be as follows:

(a) The navigation receiver, except for the ship antenna coupling unit, shall be dripproof in accordance with MIL-STD-108.
(b) The navigation receiver whip antenna coupling unit shall be watertight in accordance with MIL-STD-108.

3.3.7 Tropicalization.- Complete tropicalization of the equipment is not required and should not be used unless necessary. Insofar as possible, all material, parts, and processes used shall be a type which is nonnutrient for fungus. If a nutrient part is required, it shall be tropicalized prior to installation in the equipment.

3.3.8 Shock, vibration and inclination.- The equipment shall be capable of withstanding the shock requirements of Grade A, Class I, lightweight, type A of MIL-S-901 and the vibration requirements of type I of MIL-STD-167. The inclination requirements shall be as specified in MIL-E-16400.

3.3.9 Semiconductors.- Transistorization shall be used for those circuits for which transistor techniques are well established. Advancement of the state of the art of semiconductor circuitry is not intended and such circuitry will not be approved unless comprehensive testing by the contractor demonstrates proper performance for all probable conditions of operation. All semiconductor circuitry must be approved by the bureau or agency concerned prior to incorporation in the equipment design.

3.3.10 Controls, indicators, and panel layout.-

3.3.10.1 The number of controls and indicators shall be kept to a minimum consistent with the requirements of this specification. The requirement for control knobs of distinctive shape and color shall not apply.

3.3.10.2 Dial and vernier marking.- Where verniers are employed to subdivide dial divisions, the marking and numbering shall be such as to provide continuity throughout the dial range in accordance with MIL-E-16400.

3.3.10.3 Mechanical stability.- All controls of the equipment shall be so designed that when the equipment is properly installed and either operating or nonoperating, the setting, position, or adjustment of any control shall not be altered by any of the specified service conditions. Retention of control by locking devices is not required.

3.3.10.4 Elapsed time meters.- Time meters to indicate elapsed operating time are not required.

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3.3.10.5 Panel layouts and arrangement of operating controls.- Panel layouts and arrangement of operating controls shall be subject to the approval of the bureau or agency concerned.

3.3.10.6 Controls, indicators, and panel layouts shall conform to the requirements of MIL-E-16400, except as specified herein.

3.3.11 Listing of specifications for any given material or item herein does not in itself, constitute approval of the use of such material or item in the equipment described herein.

3.3.12 Since protective coatings may act as insulators, the requirements for such coatings shall be relaxed insofar as necessary to provide positive electrical contact, as measured by ohmmeter readings, wherever metal-to-metal contact is made.

3.3.13 Equipment design.- The equipment shall be of a modular construction with plugin modular assemblies, where feasible, in accordance with MIL-E-16400.

3.3.13.1 Modular assemblies.- The design objective is to achieve reliability, performance, and simplification of maintenance and repair through the use of modular assemblies. The modular assemblies shall be repairable on board ship. The objective for all modular assemblies and sub-assemblies is to keep the number of different units to a minimum. In the event non-repairable circuit modules or subassemblies of the throw-away type are used, the following factors shall govern:

(a) Nonrepairable circuit modules shall be designed for a reliability goal of a mean-time-between-failure of 10,000 hours. A test demonstration is not required.
(b) Each circuit module shall be treated as a single part and nonstandard part approval therefore must be obtained prior to the construction of any equipment models.
(c) Data supplied for nonstandard part approval shall include cost comparison with a repairable circuit module design as well as manufacturing information specified in MIL-E-16400 for modular assemblies.

3.3.13.2 Interchangeability.- All modular assemblies and subassemblies including plug-in circuit card types, 100 KC. R.F. Oscillator oven subassembly, and encapsultated throw-away types shall be electrically and mechanically interchangeable with those of the AN/SPN-38 and the AN/WPN-5 equipments. The interchangeability requirement 3.4.4 of MIL-E-16400 applies.

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3.3.14 Reliability.- The reliability for this equipment shall be 1000 hours mean-time-between-failures (MTBF) with a confidence level of 90 percent. The MTBF of the equipment shall be predicted using Method D of NAVSHIPS 93820. For test purposes, demonstration of attainment of a reliability level of 300 hours MTBF with a 90 percent confidence level will be required. Test level number 1 of MIL-R-22732 shall be used to determine the reliability test environment.

3.3.15 Safety requirements.- Conformance with the safety requirements specified in MIL-E-16400 shall be required.

3.3.16 Blown fuse indicators are required and shall be of the lamp indicating type.

3.3.17 All external electrical and electronic connections to electronic equipment shall be made by the use of connectors mounted in an accessible area of the external surface other than the equipment front. The connectors shall be in accordance with MIL-E-16400. Connectors selected for external electrical and electronic connections shall be compatible with the cables used in both shipboard and submarine installations.

3.3.17.1 The connector receptacles except for radio frequency connectors, shall contain the pin contacts. All connector receptacles shall be protected by a suitable cap.

3.3.17.2 Terminal boards, terminal strips and connectors.- Terminal boards, terminal strips and multicontact connectors used in interconnecting assemblies and multicontact connectors used in interconnecting major units shall have 10 percent extra unused terminals or contacts, but in no case less than two. Where different size connecting elements are utilized, the repair parts shall be approved by the Government. No repair contacts are required on multicontact connectors used for primary power connections. Stud type terminal boards shall be in accordance with MIL-T-16784. Screw type in accordance with MIL-T-55164.

3.4 Performance.- The design of the equipment shall be such that it will be capable of furnishing reliable navigation information when operated by persons having only limited training in the technical aspects of its design. The navigation receiver shall be capable of simultaneous automatic track of two Loran station pairs (i.e., master and any two slave stations). Automatic search techniques shall be incorporated such that the equipment shall successively locate and track the master while the approximate location of the slave signals is accomplished by a manual input by the operator. The equipment shall automatically "pull-in" the slave signals from the approximate location and shall automatically track both the slave signals with the master signal. The equipment shall be capable of utilizing Loran C information

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transmitted in the form of a group of eight pulses phase-coded in accordance with codes specified herein. The circuitry required for phase-coding and multi-pulsed triggering functions should be incorporated in a single chassis such that if more sophisticated and complex codes are subsequently employed modification of the equipment could be accomplished by chassis substitution. Provisions shall be provided for remote use of the two time differences being tracked by the navigation receiver as well as all necessary fail-safe indications. A visual display unit to be used as an aid in operating shall be provided.

3.5 System characteristics.-

3.5.1 Pulse shape.- The Loran C pulse shape is presently an unsymmetrical pulse having a fast rising (essentially cosine function) leading edge together with a relatively slow decay of the trailing edge. The general criteria in regard to pulse shape is that a usable amplitude be reached by 30 usec. subsequent to start of the pulse such that the ground wave pulse can be sampled at a point uncontaminated by sky waves. Further investigation of optimum pulse shape will be conducted; however, the general criteria concerning the rise time characteristics are prerequisites and will be retained.

3.5.2 Carrier frequency.- Loran C signals are transmitted on a carrier frequency of 100 kc. In accordance with international regulations, a minimum of 99 percent of the transmitted energy is confined to the band 90 to 110 kc.

3.5.3 Basic repetition rates.- Loran C signal will be transmitted on any of the specific repetition rates. Transmissions from a master and three slave stations will normally be accommodated on a single specific rate. Loran C transmissions will be accomplished using either group transmission, eight pulses/group at each slave and nine pulses/group at the master (the ninth pulse at the master is used for visual identification of the master signal).

3.5.4 Ground wave/skywave resolution.- The minimum separation between reception of ground wave and skywave signals in the service area is approximately 30 usec. Provision shall be incorporated for setting the sampling gate to approximately 20, 25, or 30 usec.

3.5.5 Loran C phase coding.- Phase coding is a technique of changing the phase relationship of the radio frequency carrier with respect to the pulse envelope in individual pulses in a pulse group. Group pulsing is employed in Loran C to raise the average power transmitted without the necessity of peak power increases. Phase coding is utilized for signal identification and to avoid contamination by sky waves of an early pulse in the group to subsequent ground wave pulses in the group. The simplified phase codes are repetitive after every two pulse groups. The same phase code is used for each slave

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transmission. The master code is of the same basic structure as the slave, except that the first four pulses of the master group are encoded in the same manner as the last four pulses of the slave group, and the last four pulses of the master are encoded in the same manner as the first four pulses of the slave. Although there are a number of acceptable combinations of this basic code structure, the following is used on all repetition rates for simplicity:

Master signal.

	Pulse-phase in degree								
Group	1	2	3	4	5	6	7	8	9
1	0	0	180	180	0	180	0	180	0
2	0	180	180	0	0	0	0	0	180

Slave signal.

	Pulse-phase in degrees							
Group	1	2	3	4	5	6	7	8
1	0	0	0	0	0	180	180	0
2	0	180	0	180	0	0	180	180

3.6 Detailed requirements.- The navigation receiver shall contain all circuits required to select, receive, detect, measure and indicate the information required in accordance with this specification when used in the Loran C system.

3.6.1 Receiver.- The equipment shall incorporate a 100-kc. radio frequency receiver.

3.6.1.1 Bandwidth.- Two bandwidths, one optimized for signal search and one for tracking, shall be incorporated. The search function selectivity shall be capable of control which will make it selectable both during normal search periods and when not operating in the search function.

3.6.1.2 Radio frequency interference rejection.- The interference rejection of the receiver without notch filters shall enable the receiver to meet the performance requirements of 3.6.3.11. In addition, two selectable calibrated notch filters of identical characteristics and separately adjustable over the band 70 to 130 kc. for suppression of interfering A1 and F1 type emissions shall be provided. These filters shall provide a 30-db suppression of an A1 or F1 signal with a minimum practical bandwidth no greater than 10 kc. at minus 6 db and 1 kc. at minus 30 db.

3.6.1.3 Overload and recovery-time characteristics.- The receiver shall be designed to have as fast a recovery time after receiver saturation as is practicable. In any event, it shall recover in not more than 50 usec. after a signal causing saturation is removed.

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3.6.1.4 Sensitivity and noise figure.- The receiver shall have a sensitivity such that it can operate with antenna specified in 3.6.2.1 in a field of 5 microvolts per meter (uv/m). The noise figure of the receiver shall be such that operation is limited only by signal to atmospheric noise level, and not by receiver noise.

3.6.1.5 Maximum signal input voltages.- The receiver shall be capable of operating properly with a maximum signal input corresponding to that which would be obtained when using the antenna specified in 3.6.2.1 in a field of 5 volts/meter. A switched fixed attenuator may be employed, if required, in order to obtain the required range of attenuation.

3.6.1.6 Signal amplitude balance.- Signal amplitude balance shall be automatically maintained over a range of amplitude unbalance of at least 10,000 to 1 (80 db) to the degree required to obtain specified performance anywhere in the range of from 5 volts/meter to 5 uv/m. when using the antenna specified in 3.6.2.1.

3.6.2 Receiving antenna and coupling units.- Suitable antenna and coupling units shall be supplied for use on surface ships or for use on submarines as specified in 3.6.2.1, 3.6.2.2, and 3.6.2.3.

3.6.2.1 Surface ship installation.- A suitable whip antenna and coupling unit shall be supplied for shipboard installation. The antenna shall be designed to minimize the effects of precipitation static and shall have a pluming suppressor (that is, a ball tip or similar device to suppress corona effects). Suitability of the antenna configurations recommended by the contractor will be determined by the bureau or agency concerned. In the design of the antenna, consideration shall be given to minimizing the differential phase shifts caused by reradiating structures. Coaxial cable up to 300 feet long will be used to interconnect the antenna coupling unit and the receiver.

3.6.2.2 Submarine installations. A suitable antenna coupling unit shall be supplied for submarine installation. The unit shall consist of a preamplifier unit housed in a dripproof enclosure and an azimuth indicator unit which shall be installed in a suitable space in the receiver. The preamplifier shall accept an input from a standard Navy type At-317(B)/BRR loop antenna through approximately 100 to 180 feet of RG-264A/U cable in accordance with RE49B740 and also from a 36 foot whip antenna connected through 100 to 180 feet of RG-17A/U cable in accordance with MIL-C-17/10 or RG-12A/U cable in accordance with MIL-C-17/7. Also two 50 ohm inputs for coupling through a universal coupler preamplifier to other antennas shall be provided. The azimuth indicator unit shall provide for selection of the correct

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antenna loop combinations. A switch shall be provided on the receiver for selection of the loop or whip antenna. A manual bearing input knot shall be provided for setting approximate station bearings. Compensation for ships heading shall be provided automatically from ships gyro information.

3.6.2.3 The preamplifier unit (for loop antenna operation) shall be designed so that the signal to noise ratio at the output of the r.f. plifier be not less than 1:1 when the loop antenna for the preamplifier is operating in a field of 35 microvolts per meter and there is no noise on the input signal. The time difference variation caused by heading effects shall not exceed 0.02 microseconds. Also the r.f. amplifier design shall be such that the output signal to noise ratio be not less than 1:1 with a 35 foot whip antenna (submarine installation) operating in a noise free 8 microvolt per meter field.

3.6.3 Frequency control and presentation.-

3.6.3.1 Oscillator.- A basic timing oscillator of size and complexity compatible with the required accuracy, stability and operational characteristics shall be provided. Particular care shall be taken to insure that the short-time stability is adequate for all specified requirements. The maximum drift of the oscillator shall not exceed 5 parts in 10^8 over a 10-minute period nor 1 part in 10^8 over a 24-hour period under normal operation. The frequency drift shall not exceed 1 part in 10^7 when the equipment is subjected to a 40°C. change in temperature within the ambient temperature range of 0° to 50°C. and with humidity variations of 0 to 95 percent. The foregoing must be considered minimum requirements and are not necessarily adequate to meet other requirements of this specification. Suitable switching provisions shall be provided to permit the oscillator heater circuits to be on when the receiver is in standby or off condition.

3.6.3.2 Manual frequency control.- A manual frequency or drift control shall be provided internal to the equipment, the use of which will enable the operator to make compensatory adjustments for slight variations in the timer oscillator frequency. This control shall permit adjustment of the crystal oscillator over a range of plus and minus 1 part in 10^6. Suitable control range shall be provided on the oscillator as a separate control to allow for crystal aging.

3.6.3.3 Automatic control of timer oscillator.- Means shall be provided such that after automatic search, the timer oscillator will be automatically locked with respect to the r.f. phase of the master pulse. It shall have a pull-in capability of at least plus and minus 6 parts in 10^7 frequency difference.

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3.6.3.4 Frequency dividers.- Frequency dividers utilized to derive the various Loran repetition rates from the crystal oscillator shall be of the highly stable type. Divider jumps shall not occur when tested in accordance with all requirements specified herein. Timing waveforms over the entire pulse repetition interval shall be continuous. Discontinuity in the timing waveforms as required to obtain specific pulse repetition intervals shall be accomplished only at the end of the interval.

3.6.3.5 Time difference measurement.- The measurement of the time difference between master and slave pulse envelopes and between master and slave r.f. phase shall be accomplished by an electronic digital means. It shall be possible to measure time delay continuously from a minimum of 5500 or 11,000 usec. to a maximum of the pulse repetition interval minus 5,000 or 10,000 usec. when operating with pulses spaced 500 or 1000 usec. respectively. It shall be possible to slew each indicated time delay 10,000 usec. within 60 seconds.

3.6.3.6 Time sharing of repetition interval.- The repetition interval will normally be shared by the transmissions from a master station and up to three slave stations. It is not possible to fix definite time or percentage lengths for the slave time intervals since these will be dicated by system deployment, repetition rate assignment, and other factors. Since it must be possible for the slave servo systems to lock-in and track any of the slave station signals regardless of their location in the total time interval, extreme flexibility shall be designed into the equipment.

3.6.3.7 Detection and strobing.- Sophisticated detection and strobing techniques will be required to obtain the performance required by 3.6.3.11. Carefully balanced circuits having a high degree of sensitivity and a large linear dynamic range will be required. Both linear detection and synchronous detection (using both quadrature and in-phase reference signals) will undoubtedly be required to obtain the accuracy required in the presence of noise and interference, identify cycles for phase comparison, furnish automatic gain control, operate various alarms and indicators, and perform automatic search and track functions. Careful attention must be paid to overall filtering performed in the critical detection and strobing portions of the equipment.

3.6.3.8 Time difference presentation.- The time difference between received pulses from the master and each of two slaves shall be indicated by direct in-line readout indicators. These indicators shall be easily read at a distance of 2 feet and at any angle up to 30 degrees from normal to the face of the indicator. Provision shall be made to prevent parallax errors in making readings. The envelope (coarse) and cycle (vernier) indicators shall be of an intergrated display type to give an overall time difference reading. A suitable indicator shall indicate correlation between envelope and cycle time difference readings and a visual alarm actuated when the indicated time difference is in error by an amount corresponding to one-half or more radio-fre-

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quency cycles (that is, when correct cycle identification has not been accomplished). The time-difference indication shall be marked in steps of 0.1 usec. and provision shall be incorporated for cross referencing the envelope and phase readings in a manner which will prevent ambiguous indications.

3.6.3.9 Receiver output.- The equipment shall have provisions for remote indication of the two time differences being tracked as well as all necessary failure indications. This shall be (1) a suitable output for a standard one milliampere movement ship chart recorder and (2) outputs with a digital computer.

3.6.3.10 Overall instrumental accuracy.- Instrumental error in time difference measurement read from the time difference dials in the ranges specified shall not exceed 0.05 usec. due to non-linearity of time difference measuring circuits and to differential signal amplitude of 80 db.

3.6.3.11 Effects of noise and interference on measurement accuracy.- For any combination of amplitude of input signals and interference conditions with which the equipment is herein specified to operate, the total error including instrumental errors and those introduced by effects of noise and nonsynchronous interference, or measurement on time difference between appropriate ground-wave pulses shall be less than those specified below for 95 percent of the observation.

Application	S/N weakest signal (db)	Amplitude differential (db)	Interfering signal frequency (kc./s.)	Weakest signal field/interfering field (db)	Microseconds error: Envelope	Microseconds error: Cycle Phase
Fixed	0	0 - 80	85 - 115	-35		0.06
Monitor,	0	0 - 80	85, 115	-45		.06
Ship, or	-20	0 - 80	85, 115	-35	4.5	.35
Submarine (maximum velocity 45 knots)	-20	0 - 80	85, 115	-45	4.5	.35

The acceleration characteristics of the receiver shall be such that it does not

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lose synchronization under the following conditions: A ship traveling along the baseline at 35 knots reversing its direction on a 1000-foot turning diameter. The equipment shall recover from the effects of the above accelerations in less than 8 minutes after reaching a constant velocity. The accuracy of the readings shall not be degraded by the accelerations due to normal pitch, roll, and yaw of the ship.

3.6.4 Search, synchronization and clear guard.- The equipment shall be capable of automatic master signal search and synchronization, and shall have facility incorporated to indicate that synchronization is on the groundwave and not on a sky wave. The clear guard shall operate properly in the event of large amplitude differentials between groundwave and sky wave signals. Facility shall be incorporated for disabling the clear guard function when intentional sky wave synchronization is intended. The equipment shall be capable of automatic search and synchronization under all of the signal amplitude conditons, signal to noise ratios, and velocities herein specified. Provisions shall be incorporated for manually slewing the search function in either direction at two rates of speed to allow setting of the search gates to the expected time difference thereby reducing search time. Search shall be capable of being accomplished using either of the selectivities provided in the receiver. When the narrow-band selectivity is used for search, provision shall be incorporated which prevents a loss of synchronization upon switching to the wide-band selectivity. Should the maximum automatic search time for signals under the worst conditions specified exceed 3 minutes, a variable speed search shall be used to allow selection of the optimum speed. The slave signals shall automatically pull in to synchronization when the equipment is set manually to within 100 microseconds of the low side of the correct reading for each of the two slave stations.

3.6.5 Indicator lights.- The following indicators shall be provided in the form of front panel lights on the receiving set. Provision shall be made for remote presentation of all alarms and indicators at computers, recorders, and other accessory equipment which may be used with this equipment:

(a) Loss of any of the three Loran C signals, identifiable by selector switch.
(b) Blink indicator.
(c) Delay range of any of the servo units has reached its limit.
(d) Synchronization on sky wave, alarm actuated by clear guard.
(e) Synchronization lock-in indicator operating in accordance with the position of the search selector switch.
(f) Power operate and power standby light.
(g) Narrow band alarm.
(h) Acceleration error indicator.

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3.6.6 Blink.- A visual alarm shall be provided which will operate reliably at all signal-to-noise ratios down to 1 to 10. The blink procedures for Loran C ground station equipment are as follows:

(a) Master station 9th pulse transmission.- The 9th pulse shall occur 2000 microseconds after the 8th pulse when 1000 microsecond spacing is used. A code, which the receiver need not recognize shall be transmitted by the master station to signal the offending slave station. During periods of blink the 9th pulse shall be in its normal position approximately 2 seconds out of every 10 seconds.

(b) Slave station blink transmission.- All slave stations shall have a common blink procedure. The blink shall consist of the first two of the eight pulses blinking on and off. The duration of the "on" portion shall be between 0.2 and 0.35 seconds and shall repeat every 4 seconds. The remaining 6 of the 8 pulses shall be continuously "on" with the normal phase code.

3.6.7 A visual display cathode-ray type, shall be provided with appropriate slow and fast sweep speeds, to permit operational observation of r.f. Loran C signals. For the operational function two continuously and separately adjustable calibrated notch-type filters, each of which are continuously adjustable from 70 to 130 kcs. and which can be applied to filter the visual r.f. presentation without affecting the signal voltages delivered to the synchronous detector of the synchronizer, shall be provided. (These are in addition to and shall be similar to the notch filters required for the radio receiver in 3.6.1.2.) In addition, a selectable band pass filter which has a pass band considerably smaller than that required for system performance, shall be provided to improve the visual signal-to-noise ratio for signal search. A pass band of 2 kc. is suggested for this filter although the contractor may propose another figure, which can be demonstrated as optimum, in consideration of the length of the Loran C pulse and the average shapes of the atmospheric noise pulses at 100 kc.

3.6.7.1 The visual display shall consist of a 3-inch rectangular cathode-ray tube and the necessary circuits to perform the functions specified herein.

3.6.7.2 Operational display.- For the purpose of assisting the initial positioning of Loran C signals for automatic operation, the visual display shall display selectable presentations as outlined in the following paragraphs. Consideration should be given to techniques for substitution of visual signal search for automatic search in the presence of poor signal to noise ratios.

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3.6.7.2.1 Horizontal sweeps.- Two sweeps (full oscilloscope capability). Provision shall be made for selection of three sweep speeds. Sweep lengths shall be as shown below; however, fixed optimum selectable sweep lengths may be provided subject to approval by the bureau or agency concerned.

Sweep Speed	Sweep length (usec.)
Fast	250 - 1250
Medium	9000
Slow	1/2 recurrence interval

The slow sweep is triggered by the master; the top fast and medium sweeps are triggered by the master; the bottom fast and medium sweeps are triggered by slave A or B.

3.6.7.2.2 Vertical deflections.- The vertical deflections shall be as follows:

(a) Loran C r.f. signals, master and one slave at a time.
(b) Same as (a) except filtered as described in 3.6.7.
(c) Presentation to show coding phase.
(d) Automatic gain control, master and two slaves, for use in indicating relative signal strengths.

3.6.7.3 A front panel meter shall be provided to permit operational observation of the internally-generated control and error signals. This meter shall perform the following functions as a minimum:

(a) Display automatic gain control voltages of the receiver master and slave channels separately for use in indicating relative and absolute signal strengths.
(b) Indicate envelope-to-cycle discrepancy unless this indication is provided elsewhere.
(c) Monitor frequency drift of timer-oscillator.

3.6.8 Testing.- For the purpose of facilitating the alignment and servicing of the receiving set, the visual display and front panel meter shall be provided with probes for temporary connection to the probe points in the set and a probe for temporary connection to the appropriate horizontal sweep trigger probe points in the set, together with the vertical gain controls, sweep speed and amplitude controls, trigger polarity reversing switch and other facilities as are found necessary in accordance with the design features. Facility shall

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be provided for connecting the probes to the visual display and meter inputs and storage space shall be provided for them within the equipment.

3.6.8.1 Test points.- Test points shall be in accordance with MIL-STD-415 and shall be isolated to prevent degradation in operation when used with suitable test probes, and shall be provided for the following:

(a) Monitoring receiver r.f. output, phase detector output and video pulses.
(b) Monitoring outputs of frequency generator and phase circuitry.
(c) Monitoring stability of the outputs of all pertinent gate and trigger generating circuitry.
(d) All test points required for receiver indicator calibration and alignment.

Test points at which suitable sweep triggers can be obtained shall be provided for all of the above test points. Should a noise sampling circuit be employed in the equipment, a suitable test point from which an indication of the noise amplitude can be obtained shall be provided.

3.6.9 Serviceability.- Each unit of the equipment shall be serviceable in its mounted position. A simple slide track arrangement shall be provided to facilitate withdrawal of the panel chassis assembly. A locking device and safety catch shall be provided to hold the panel chassis assembly in the withdrawn position. Complete accessibility of all units shall be provided for by tilting of the panel chassis and its subassemblies. Locking devices shall be provided for all tilt facilities. When the panel chassis and subassemblies are in the servicing position they shall not obstruct equipment which is mounted on either side. The equipment shall be serviceable without the use of extension cables, test jigs, or plug-in adapters. The use of plug-in subassemblies must be approved by the bureau or agency concerned prior to incorporation in the equipment design. Maintainability requirements shall be in accordance with MIL-M-23313. The equipment repair time (ERT) shall not exceed 15 minutes. A preproduction maintainability test report is required.

3.6.10 Signal simulator.- A signal simulator and special test equipment is intended to simulate the transmission of master and slave signals in the Loran C system and to facilitate the checking, calibration, maintenance and repair of the equipment on board ship. These test facilities are not intended to duplicate available shipboard test equipment but to supplement them and to enable checking and servicing the equipment without dependence upon on-the-air signals. These test facilities shall be self contained in the equipment.

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3.7 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the cognizant government inspector for system approval prior to preproduction inspection. Any changes to the approved quality assurance system which might affect the degree of assurance required by this specification or other applicable documents must be submitted to the cognizant inspector and approved in writing prior to use.

4.1.2 Government Verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Sampling, examination, and test procedures shall be in accordance with MIL-E-16400, except as otherwise specified herein.

4.3 Scope of tests.- The tests specified herein are minimum requirements. Additional tests may be requested by the Government inspector or by the bureau or agency concerned, if determined to be required to demonstrate that the equipment is constructed and performs as required by this specification. Prior to initiating any tests the contractor shall submit to the bureau or agency concerned via the Government inspector five copies of detailed test procedures for all test categories. The detailed test procedures shall reference the requirements of this specification by paragraph number as an indication of the function being tested. Simplified block diagrams which indicate the test

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setup and the types of test instruments to be used shall also be included in the detailed test procedure. The approved tests shall form a part of this specification and shall be utilized by the Government inspector as a basis for inspection. The tabulated data on all tests shall be submitted to the Government inspector and the test sheets shall also function as data sheets for notation of observed performance.

4.4 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in 4.4.1 through 4.4.4 and 4.5.3.

4.4.1 The preproduction equipment shall be subjected to all the preproduction inspection of MIL-E-16400, with the exception of the accelerated life test. The preproduction inspection shall specifically include all tests necessary to show compliance with the provisions of this specification with emphasis on all paragraphs under 3.6.

4.4.2 Detailed requirements.- The equipment shall be tested to determine performance under real or simulated conditions of expected man-made interference, natural external noise, movement of the navigation equipment with respect to the shore stations and interference from other signals. Except where tolerances are specifically stated, the detailed design characteristics and performance specifications are minimum requirements. After initial adjustments, prior to or during tests, as applicable, the equipment characteristics and performance shall remain stable without readjustment of any other than the operator's controls shall only be required periodically in a manner consistent with his normal operational capability.

4.4.3 In lieu of the accelerated life test specified in MIL-E-16400, a program of tests to establish verification of the MTBF and reliability requirements specified in 3.3.14 shall be established and maintained in accordance with MIL-R-22732.

4.4.4 Field operating tests.- In addition to the preproduction inspection the equipment, at the discretion of the bureau or agency concerned, shall be subjected to a field operating test with the equipment installed at sites or on ships to be designated or provided by the bureau or agency concerned. These tests will not be performed by the contractor. Special test equipment or simulators which have been built by the contractor in support of this equipment shall be made available to the bureau or agency concerned during these tests. It is anticipated that the field operating tests of the preproduction receiver will extend over a period of from 30 to 60 days. The preproduction model shall be monitored by the assigned inspector and verified by him as to its readiness for field tests before it is submitted to the bureau or agency concerned for field operating tests. If the field tests indicate the necessity of returning the model to the contractor for modification, it shall be resubmitted to the inspector for his verification of readiness before returning it to the bureau or agency concerned for completion of the field tests.

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4.4.5 Preproduction approval.- Final approval of the preproduction equipment, based on successful completion of all tests will be furnished the contractor in writing by the bureau or agency concerned. After approval, no changes shall be made in the circuitry, parts, or mechanical design of the equipment, except by written authority of the bureau or agency concerned. The official approval of the preproduction model must be received by the contractor before production of the remaining equipments will be authorized.

4.4.6 Test failures.- Failure of the equipment to pass a specific test will be reported immediately to the bureau or agency concerned. The test failure report shall give complete and comprehensive information on the nature of the failure and recommended corrective measures. Failures of electron tubes are not to be considered as equipment failures, unless the tube failure is caused by improper application of the tube in the equipment.

4.5 Quality conformance inspection.-

4.5.1 Production inspection.- Production inspection, in accordance with MIL-E-16400, shall be performed on all production equipments.

4.5.2 Production control inspection.- Production control inspection in accordance with MIL-E-16400, shall be performed on equipments selected at random in accordance with the following sampling plan:

Total quantity of production	No. of samples
29 or less	1
30 to 40	2
41 to 80	3

4.5.3 Reliability testing.- Reliability testing shall be performed in accordance with test level 1 of MIL-R-22732.

4.5.3.1 Demonstration of the required reliability level of 1000 hours MTBF shall be accomplished by demonstrating a minimum MTBF of 300 hours with 90 percent confidence. Accept-reject criteria shall be in accordance with the accept-reject criteria specified in MIL-R-22732.

4.5.3.2 Reliability testing shall be conducted on the preproduction equipment and each production control equipment.

4.5.3.3 Performance parameters to be monitored during reference testing shall be proposed by the contractor and for bureau or agency concerned approval.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Basic equipment or item.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing, and marking.- The equipment, accessories, technical publications, and repair parts shall be preserved and packaged by level A or C, packed by levels A, B, or C and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification, and the type of equipment.
(b) Level of packaging, packing, and marking required.

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6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirements for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5825-N080Sh)

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